

UK Road Traffic Accident Severity Prediction — Project Report

1. Summary

This project analyzes UK road traffic accident data (STATS19, 2018) to build a predictive model that identifies accidents likely to result in a **Killed or Seriously Injured (KSI)** outcome. By combining accident-level data (location, time, weather, road type) with vehicle-level data (driver age, vehicle count, engine size), we built a binary classification pipeline — from data cleaning through model tuning — and translated the results into practical road-safety recommendations.

2. Problem Statement

The challenge: Road traffic accidents in the UK vary widely in severity — from minor collisions to those resulting in serious injury or death. Emergency services, city planners, and road-safety authorities currently respond to accidents largely after they happen, with limited ability to predict where and under what conditions a severe accident is more likely to occur.

The question: Can we predict, from the circumstances of an accident (location, time, weather, road type, speed limit, vehicles involved), whether it is likely to result in a KSI outcome — before resources are allocated?

Why it matters:

- Road-safety budgets and police patrols are limited and must be prioritized
- Most existing responses are reactive, not predictive
- Identifying high-risk patterns in advance allows for preventive, not just corrective, action

Our approach: framed as a binary classification problem.

- Target: `is_KSI` — 1 if the accident is Fatal or Serious, 0 if Slight
- Input: accident-level features (time, location, weather, road type, speed limit) combined with vehicle-level features (driver age, vehicle count, engine size)

- Goal: a model that flags high-risk accidents/conditions with enough recall to be useful for real-world safety planning — not just statistically accurate, but usable for action

3. Data Sources

- **Accidents_2018.csv** — 122,635 accidents, one row per accident: location, time, weather, road type, speed limit
- **Vehicles_2018.csv** — 226,409 vehicle records, one row per vehicle involved in an accident (an accident can involve multiple vehicles)
- **Casualties_2018.csv** — 160,597 casualties — deliberately excluded from modeling (see Section 6)

All three are official STATS19 records published by the UK Department for Transport.

4. Data Merging

Since the relationship between accidents and vehicles is one-to-many, vehicle data was first aggregated to one row per accident before merging:

- `veh_count`: number of vehicles in the accident
- `avg_driver_age`: average driver age
- `max_engine_cc`: largest engine capacity among the involved vehicles
- `avg_vehicle_age`: average vehicle age
- `n_male_drivers`: number of male drivers

The `Accident_Index` key was cast to string in both datasets to resolve a mixed-type warning before merging. The merge used `how='left'` (keep all accidents) and `validate='one_to_one'` (fail loudly if any duplicate keys remained after aggregation), producing a combined dataset of 122,635 accidents.

5. Data Cleaning

- Aggregated the one-to-many Vehicles table to accident level before merging.
- Dropped rows missing coordinates/time (about 0.05% of data — negligible).
- Dropped `LSOA_of_Accident_Location` (high cardinality, redundant with Lat/Long) and `Number_of_Vehicles` (redundant with the aggregated `veh_count`).
- Converted -1 ("unknown") codes to NaN before any calculations, to avoid distorting averages.

- Fixed data-entry errors: max_engine_cc above 10,000 and avg_driver_age below 15 set to NaN (raw engine-cc histogram showed impossible values up to about 100,000), then imputed using DecisionTreeRegressor.
- Checked for duplicates — none found.
- After cleaning, max_engine_cc settled into a realistic 1,500–2,000cc peak; avg_driver_age stayed centered around 30–40.

6. Data Leakage Prevention

Data leakage — information that would not realistically be available at prediction time — was addressed at several levels:

1. **Casualties dataset excluded entirely** from the merge, since fields like Casualty_Severity are near-duplicates of the target and would let the model "see the answer."
2. **Accident_Severity removed from features (X)** — since the target is_KSI is directly derived from it, leaving it in would let the model learn a trivial rule instead of a real pattern.
3. **Did_Police_Officer_Attend_Scene_of_Accident removed** after appearing with suspiciously high feature importance — police attendance is plausibly a *consequence* of severity (severe accidents are more likely to be attended) rather than a cause available before the outcome is known. Two other near-zero-importance features (Junction_Control, 2nd_Road_Class) were dropped at the same time.
4. **Train/test separation enforced throughout preprocessing** — imputation, target encoding, and scaling were all fit only on the training set and only applied to the test set, preventing information from the test set leaking into training.

7. Feature Engineering

- **Date/time decomposition:** Hour, Month, and Is_Weekend extracted from the raw date/time fields.
- **Missing-value flags** (driver_age_missing, engine_cc_missing, vehicle_age_missing) kept as binary indicators, since missingness itself correlated with severity — it wasn't random (MNAR), so it carried signal worth preserving even after imputation.
- **IterativeImputer** (using a Decision Tree estimator) used instead of simple mean/median imputation, since missingness was non-random and better predicted from other columns.

- **Target Encoding** applied to Local_Authority_(Highway) (a high-cardinality categorical column) with smoothing to avoid overfitting to rarely occurring areas.
- **Two feature sets prepared:** a scaled version (for distance-based models like Logistic Regression) and a raw/unscaled version (for tree-based models like Random Forest and XGBoost, which don't require scaling).
- Train/test split: 80/20, stratified on is_KSI (class imbalance: about 20% KSI vs 80% Slight).

8. Exploratory Data Analysis — Key Findings

- **Time of day:** KSI rate follows a smooth daily wave — lowest around 8 AM, highest around 12–4 AM.
- **Day of week:** Weekends show a higher KSI rate than weekdays — a "weekend effect" from higher speeds and leisure driving.
- **Month:** Small seasonal variation, slightly higher rates in summer.
- **Speed limit:** Non-linear — KSI rate rises steadily from 20 to 60 mph, then drops at 70 mph. Motorways are engineered to a higher safety standard despite the higher speed, which is why the raw correlation with speed limit was weak.
- **Urban vs. rural:** Urban areas have more raw accidents but a lower KSI rate; rural accidents are less frequent but more severe.
- **Lighting, weather, road surface:** Highest KSI rates cluster in conditions that reduce visibility or control — darkness, fog, flooded roads, single carriageways.
- **Road type:** Single carriageways and more complex/uncontrolled road types show higher KSI rates.
- **Geography:** A 20,000-accident hotspot map showed accident density concentrated in urban centers, but KSI vs non-KSI points were spatially intermixed — raw lat/long carries little extra signal beyond Urban_or_Rural_Area.
- **Correlations:** All individual numeric correlations with is_KSI are weak — severity is driven by non-linear, categorical interactions, not simple linear relationships.

9. Modeling and Experiments

Framing: binary classification (is_KSI), evaluated via 5-fold stratified cross-validation and a held-out test set (Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, plus train/test F1 gap to check overfitting).

The final model came from a sequence of experiments — 18 in total — each one changing something (features, data balance, hyperparameters, or approach) and checking the effect.

Stage 1 — Baseline: 5 models on the full feature set (class-weighted)

#	Model	F1	ROC-AUC	Notes
1	Logistic Regression	0.391	0.659	scaled, class_weight='balanced'
2	Decision Tree	0.390	0.646	class_weight='balanced'
3	Random Forest	0.376	0.672	200 trees, highest accuracy (0.722), lowest recall
4	XGBoost	0.413	0.685	scale_pos_weight from class ratio
5	LightGBM	0.410	0.684	same balancing approach

Stage 2 — Data leakage found and fixed

Did_Police_Officer_Attend_Scene_of_Accident dominated feature importance — dropped along with two weak features. XGBoost re-trained:

#	Model	F1	ROC-AUC
6	XGBoost (leakage removed)	0.380	0.648

Score dropped slightly — expected, since the earlier score was inflated by the leaking feature. This is the trustworthy result that goes forward.

Stage 3 — Hyperparameter tuning (RandomizedSearchCV, scored on average_precision, 5-fold CV)

#	Model	F1	ROC-AUC	PR-AUC
7	XGBoost (tuned)	0.411	0.686	0.354
8	LightGBM (tuned)	0.409	0.685	0.353

Best XGBoost params: n_estimators=300, max_depth=6, learning_rate=0.03, min_child_weight=10, subsample=0.8

XGBoost (tuned) was selected as the final model — best ROC-AUC and PR-AUC overall.

Stage 4 — Decision threshold selection

Rather than the default 0.5 cutoff, the threshold was tuned deliberately:

- F1-optimal threshold: 0.484 → F1 about 0.41, but recall only about 60–65%
- **Chosen threshold: 0.432**, tuned to guarantee at least 75% recall (Precision about 0.277, Recall about 0.750)

The recall-focused threshold was chosen over the F1-optimal one because in a road-safety context, missing a real severe accident (false negative) is more costly than an extra false alarm (false positive).

Stage 5 — Supplementary/exploratory analysis

#	Technique	Finding
9	K-Means (k=2)	No meaningful alignment with KSI vs non-KSI
10	PCA (3 components)	~25% variance explained; no clear visual separation — confirms non-linear boundary
11	One-Class SVM	Anomaly-detection alternative; not adopted for production

Stage 6 — Undersampling experiment

RandomUnderSampler balanced the training data; all 5 models re-run.

#	Model	F1	ROC-AUC
12	XGBoost (undersampled)	—	—
13	Logistic Regression (undersampled)	0.392	0.659
14	Decision Tree (undersampled)	0.382	0.634
15	Random Forest (undersampled)	0.405	0.679
16	LightGBM (undersampled)	0.359	0.681

Results were broadly comparable to class-weighting — no material improvement — so class weighting was kept for the final model.

Summary: 5 core algorithms, run across 6 stages, for 18 total experiments before arriving at the final model: **XGBoost (tuned), threshold 0.432**.

Artifacts saved:

- Model file: model_artifacts/ksi_xgboost_final.pkl
- Threshold config: model_artifacts/threshold_config.json
- Full comparison table: model_artifacts/model_comparison.csv

10. Why the Model's Performance Looks Modest

We intentionally removed leakage-related features and kept only information that would be realistically available for prediction. The target is also imbalanced, and accident severity depends on multiple interacting factors — so the problem is inherently difficult, and a modest but honest score is preferable to an inflated, leaky one.

The biggest challenges throughout the project were **data leakage** and **class imbalance**. We first detected that some casualty-related and police-attendance information could reveal the accident outcome, so we removed it. We then handled the imbalance using class weighting, threshold tuning, and under sampling, and compared several models before settling on the final approach.

11. Business Insights and Recommendations

1. Speed Management (Based on: KSI rate by speed limit; Speed limit feature importance)

- Identify high-risk speed-limit zones from KSI-rate patterns
- Increase enforcement and speed monitoring (cameras, radar) in flagged zones
- Review and adjust speed limits where risk is consistently elevated

2. Night Safety (Based on: KSI rate by hour and lighting conditions)

- Improve lighting in high-risk areas
- Increase night patrols during peak-risk hours
- Improve road visibility and lane markings for low-light conditions

3. Rural and High-Risk Roads (Based on: KSI rate by road type/urban-rural split; geographic hotspots)

- Prioritize safety audits for roads and hotspots with highest KSI concentration
- Improve road design and signage on rural roads and complex junctions
- Allocate emergency-response resources based on mapped risk density

4. Weather and Road Conditions (Based on: KSI rate by weather and road-surface conditions)

- Use weather/road-surface risk patterns to trigger real-time driver warnings
- Improve drainage and road maintenance in weather-sensitive locations
- Apply targeted safety measures during high-risk weather periods

5. Driver and Vehicle Factors (Based on: driver age, vehicle count, driver sex — confirm against actual feature importance)

- Target awareness campaigns toward driver-age groups most represented in severe accidents
- Study multi-vehicle collision patterns to inform junction and interchange design

6. Data-Driven Government Decisions (Based on: predicted KSI probability, tuned decision threshold)

- Rank locations by predicted KSI probability, not just a yes/no label
- Prioritize limited road-safety budgets toward the highest-ranked locations
- Use the model as a decision-support tool alongside expert judgment, not a replacement for it

We recommend using the model as a **risk-prioritization tool** rather than a standalone decision-maker. The government can use the predicted KSI risk to prioritize road safety interventions, enforcement, infrastructure improvements, and emergency planning.

12. Limitations

- Findings represent associations, not proven causation — any proposed intervention should be piloted before full rollout.
- The model is trained on a single year (2018) of UK data — it should be revalidated before use on other years or regions.
- The recall-focused threshold trades off precision (about 28%) — expect more false alarms in exchange for catching more real severe accidents; appropriate for a screening tool, not automated action without human review.
- The model should support, not replace, expert, and field judgment.

13. Conclusion

This project demonstrates a complete, leakage-aware pipeline for predicting road accident severity — from careful data cleaning and feature engineering through model comparison and a deliberately chosen, safety-oriented decision threshold. The resulting model and its outputs can support targeted, evidence-based road-safety interventions rather than reactive, after-the-fact responses.